Category C - Coding Assignment

Problem 10: Busiest Bus Row - Solution & Implementation

TASK DESCRIPTION

A college bus records how many students sit in each seat as a 2D grid, where each row is one row of seats. The transport office wants to know which row carries the most students. Implement a function busiestRow(grid) that returns the index of the row with the largest total and that total. If two rows tie, return the one with the smaller index.

CONSTRAINTS & EXAMPLE

- 1 <= rows, columns <= 10 | - 0 <= grid[i][j] <= 3
Example: grid = [[2, 0, 1], [3, 3, 1], [1, 1, 1]] -> **Expected Output:** Row 1, Total 7

PYTHON SOLUTION IMPLEMENTATION

```
def busiestRow(grid):
    max_total = -1
    best_row_index = 0
    for i in range(len(grid)):
        current_row_total = sum(grid[i])
        if current_row_total > max_total:  # Strict inequality preserves smaller index on ties
            max_total = current_row_total
            best_row_index = i
    return best_row_index, max_total

# Example Test Case
grid = [[2, 0, 1], [3, 3, 1], [1, 1, 1]]
row_idx, total = busiestRow(grid)
print(f"Row {row_idx}, Total {total}")  # Output: Row 1, Total 7
```

COMPLEXITY ANALYSIS

- **Time Complexity: O(r * c)** - Where r is rows and c is columns. We traverse every element in the 2D grid once.
- **Additional Space Complexity: O(1)** - Uses only scalar variables (max_total, best_row_index) for tracking.